LAB 09

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import numpy as np
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt


data = fetch_olivetti_faces(shuffle=True, random_state=42)
X = data.data
y = data.target


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)


gnb = GaussianNB()
gnb.fit(X_train, y_train)


y_pred = gnb.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy * 100:.2f}%')


print("\nClassification Report:")

print(classification_report(y_test, y_pred, zero_division=1))
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cross_val_accuracy = cross_val_score(gnb, X, y, cv=5)
print(f'\nCross-validation accuracy (Average): {cross_val_accuracy.mean() * 100:.2f}%')


fig, axes = plt.subplots(3, 5, figsize=(12, 8))
for i, ax in enumerate(axes.ravel()):

    ax.imshow(X_test[i].reshape(64, 64), cmap=plt.cm.gray)



    title_color = "green" if y_test[i] == y_pred[i] else "red"
    ax.set_title(f"True: {y_test[i]}\nPred: {y_pred[i]}", color=title_color, fontsize=10)
    ax.axis('off')

plt.tight_layout()
plt.suptitle("Face Recognition Results (Naive Bayes)", fontsize=16, y=1.05)
plt.show()
```